

Launch one EC2 using Amazon Linux 2 image and add a script in user data to install Apache.

Instance state = running X Clear filters ▾ < 1 >

☒	Name	Instance ID	Instance state	Instance type	Status check	Availability
☒	Apache-server	i-07acae39c8617bb4d	Running	t3.micro	3/3 checks passed	us-east-1c

Instance ID
 i-07acae39c8617bb4d (Apache-server)

Current user data
User data currently associated with this instance.

```
#!/bin/bash
yum update -y
yum install httpd -y
systemctl start httpd
systemctl enable httpd
```

← → ↺ ⚠ Not secure 54.90.72.22

It works!

Launch one EC2 using Ubuntu image and add a script in user data to install Nginx.

Recents Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Browse more AMIs
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

User data - optional [Info](#)
Upload a file with your user data or enter it in the field.

Choose file

```
#!/bin/bash
apt update -y
apt install nginx -y
systemctl start nginx
systemctl enable nginx
```

Copy the public ip address

	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone
✓	Nginx-server	i-0f34945ed275dcf58	Running	t3.micro	Initializing	us-east-1c

i-0f34945ed275dcf58 (Nginx-server)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Instance summary [Info](#)

Instance ID

✓ Public IPv4 address copied

Public IPv4 address

Private IPv4 addresses

in the browser run `http://publicip`



Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Launch one Windows server and install Tomcat on Windows.

In web browser

[RDP \(Remote Desktop Protocol\)](#)

How to connect

1. Open or install a remote desktop client
2. Download the RDP shortcut file
3. Open the RDP client and run the shortcut file
4. Retrieve password
5. Enter username and password
6. Connect

RDP shortcut file
[Download](#)

Public DNS

[ec2-184-72-166-38.compute-1.a](#)

Username

Administrator

Password

[Get Windows password](#)

either upload your private key file or copy and paste its contents into the field below.

Upload private key file

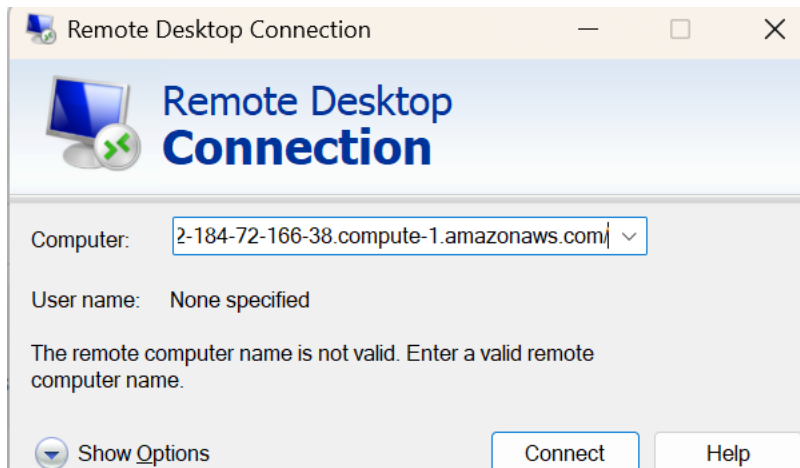
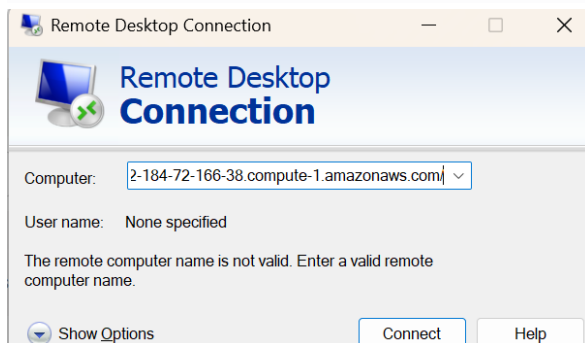
Amazon-Linux_key.pem
1.68 KB

Private key contents

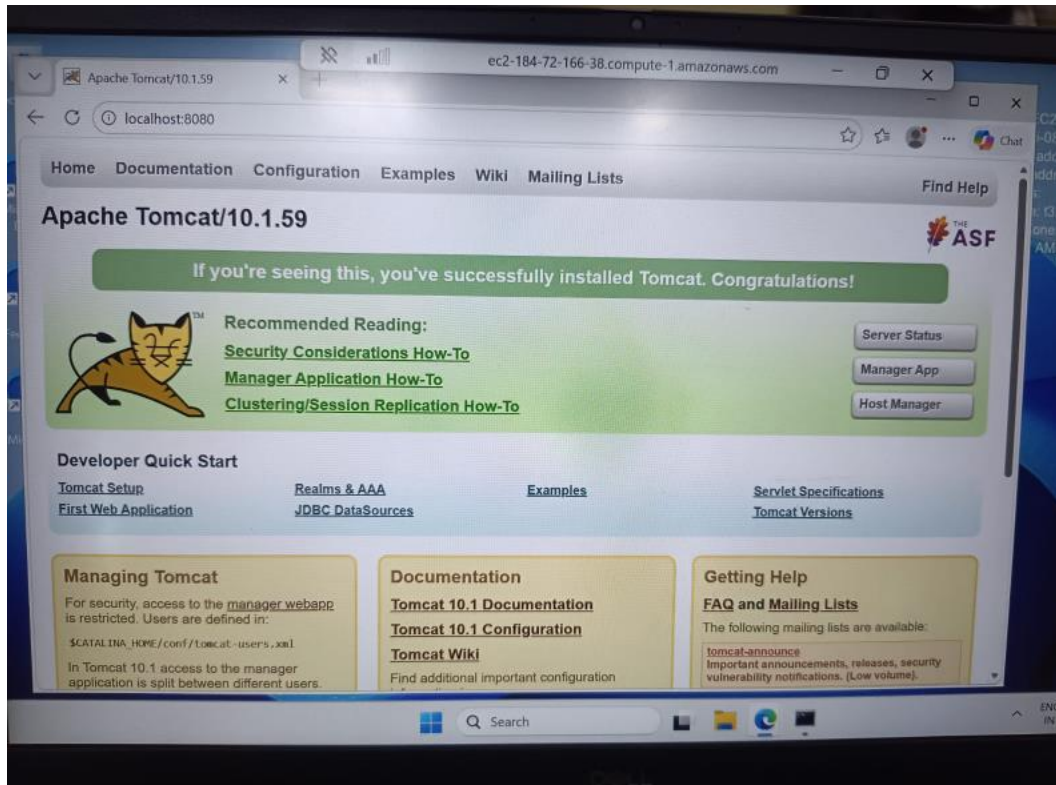
```
ED4/vJGAHLr46OnJHL2FZUof1L2GbmwHaS7H2uQQRMIeIRCLNytFuiwr2Wd3DALy
9X1o+M7e4L+iZVO1Uv3VmtECgYEAtSek2E5tWiYelOulztUZ2o0zheW2iLaO1UTy
ZXIfypVC78f2i+0Elgj8I4TAAKRREt583hg+AeKy0qTEXHXhn9mSbmW6YlvZVj8t
7JLQuGvxpcOOCg/dsB636MYP8MbJzwLg3BAJJ9ep7Zo8i/O2duFihHilU7BamBqf
mqExSNECgYEAytrbDR2m5K8pG/3XPxE60Z5AowEJ1PFQ7Pr31a5Soldp5jeayHM
wJbCGWQGMi1EZadkGB343ZJEETddVfSoQofftR5YUZE6EuQz2BCVWMkPDZFnEG/
8qD2kN04jwCni8FfucPtq9c6W1n4Mv44gk037cvM7u7cicraziolssE=
-----END RSA PRIVATE KEY-----
```

Cancel

Decrypt password



in the windows server go to microsoft edge and search <http://localhost:8080>



Tomcat is installed and working on your windows ec2 server

Take a snapshot of the instance created in Task 1.

Click the volume ID vol-07566bf1d5a340d1b.

This will open the Volumes page.

Volumes (1) [Info](#) Last updated 1 minute ago [Recycle Bin](#) [Actions](#) [Create volume](#)

Saved filter sets [Choose filter set](#)

[Volume ID = vol-07566bf1d5a340d1b](#) [Clear filters](#)

☐	Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID
☐		vol-07566bf1d5a340d1b	gp3	8 GiB	3000	125	snap-0bc9c

Click Actions at the top-right.

Then look for Create snapshot and click it.

Instance	Instance ID	State	Actions	Instance Type	Availability Zone
☒ Nginx-server	i-0f34945ed275dcf58	Running		t3.micro	us-east-1c
☐ Tomcat-windows	i-086a281520556d911	Stopping		t3.micro	us-east-1c

connect to the nginx server instance and use command whoami

```
ubuntu@ip-172-31-20-246:~$ whoami
ubuntu
ubuntu@ip-172-31-20-246:~$
```

i-0f34945ed275dcf58 (Nginx-server)

```
ubuntu@ip-172-31-20-246:~$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/ubuntu/.ssh/id_ed25519):
Enter passphrase for "/home/ubuntu/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ubuntu/.ssh/id_ed25519
Your public key has been saved in /home/ubuntu/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:rKPIDQz1/w2ez6H6JqRDIykzuY0IJp+as111zkGzITc ubuntu@ip-172-31-20-246
The key's randomart image is:
+--[ED25519 256]--+
```

```
ubuntu@ip-172-31-20-246:~$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIF4zZWVver247K+/F/b4qKGfBo5hQB4FyJg4k6NoGALU ubuntu@ip-172-31-20-246
ubuntu@ip-172-31-20-246:~$ cat ~/.ssh/id_ed25519.pub >> ~/.ssh/authorized_keys
ubuntu@ip-172-31-20-246:~$ chmod 600 ~/.ssh/authorized_keys
```

```
ubuntu@ip-172-31-20-246:~$ ssh localhost
The authenticity of host 'localhost (127.0.0.1)' can't be established.
ED25519 key fingerprint is: SHA256:K/zUknGrT/4hI91lCmW94rn4lxwMBTKilQsM6IkJifk
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

now exit from that and use command ssh localhost again, this time it doesnt ask for any password.

```
ubuntu@ip-172-31-20-246:~$ ssh localhost
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Aug 31 18:12:09 UTC 2026

System load:  0.15           Temperature:   -273.1 C
Usage of /:   34.0% of 6.61GB Processes:     123
Memory usage: 26%           Users logged in: 0
Swap usage:   0%            IPv4 address for ens5: 172.31.20.246
```

Launch any EC2 using the spot purchasing option.

Go to EC2 → Instances → Launch instances.

For now, just open the Launch an instance page.

You can use Amazon Linux 2023 and a small instance such as t3.micro.

AWS's current console puts the Spot setting under Advanced details → Purchasing option → Request Spot Instances.

Purchasing option | [Info](#)

- ☐ None
- ☐ Capacity Blocks
Launch instances for your active capacity blocks
- ☐ Interruptible Capacity Reservations
Launch instances for your interruptible Capacity Reservations
- ☒ Spot instances
Request Spot Instances at the Spot price, capped at the On-Demand price

✓ **Success**
Successfully initiated launch of instance ([i-0290344834c25d6e6](#))

► [Launch log](#)

Enable termination policy on the EC2 created in Task 2.

The screenshot shows the AWS Management Console 'Instances' page. A table lists three instances: 'Nginx-server' (ID: i-0f34945ed275dcf58), 'spot-instance' (ID: i-0290344834c25d6e6), and 'Tomcat-windows' (ID: i-086a281520556d91). The 'Nginx-server' instance is selected. A context menu is open over this instance, showing options under 'Protection and lifecycle management', with 'Change termination protection' highlighted. To the right, the 'Actions' dropdown menu is also open, showing 'Instance settings' as the selected option. Below the instance list, the 'Details' tab for the 'Nginx-server' instance is active. A modal dialog titled 'Change termination (deletion) protection' is displayed in the foreground. The dialog explains that enabling termination protection prevents accidental deletion and provides a link to 'Learn more'. It shows the 'Instance ID' as 'i-0f34945ed275dcf58 (Nginx-server)' and the 'Termination protection' status as 'Enable' with a checked checkbox. At the bottom of the dialog are 'Cancel' and 'Save' buttons.

Instances (1/5) [Info](#) Last updated 1 minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Saved filter sets
Choose filter set ▼

Find Instance by attribute or tag (case-sensitive)

	Name	Instance ID
☒	Nginx-server	i-0f34945ed275dcf58
☐	spot-instance	i-0290344834c25d6e6
☐	Tomcat-windows	i-086a281520556d91

i-0f34945ed275dcf58 (Nginx-server)

[Details](#) Status and alarms Monitor

Change termination (deletion) protection

To prevent your instance from being accidentally deleted, you can enable termination protection for the instance. [Learn more](#)

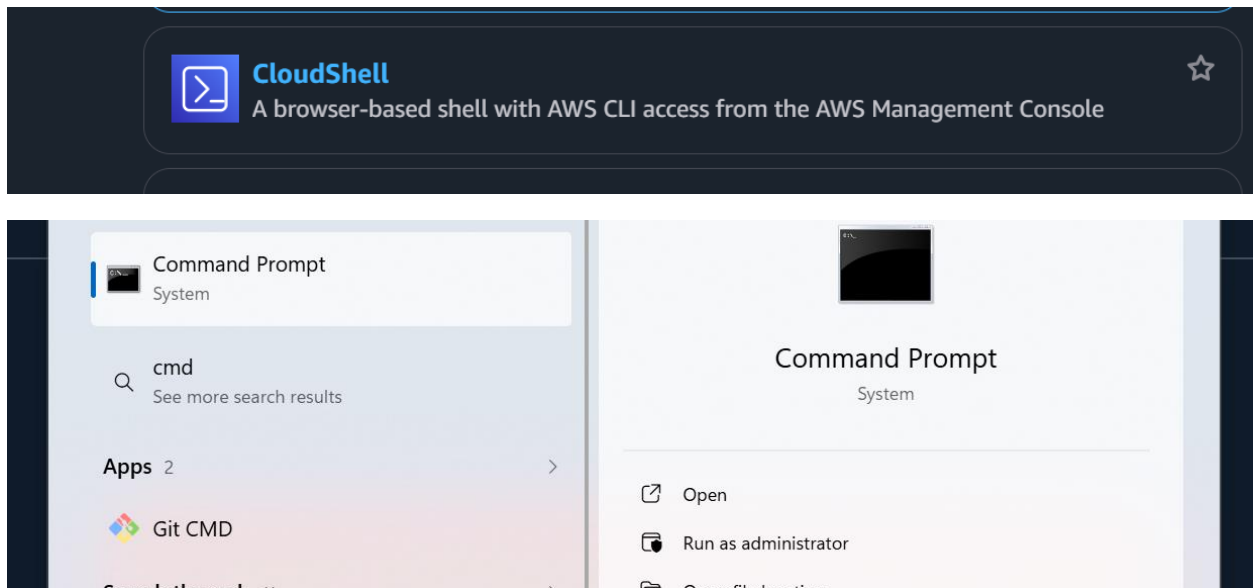
Instance ID
[i-0f34945ed275dcf58](#) (Nginx-server)

Termination protection
☒ Enable

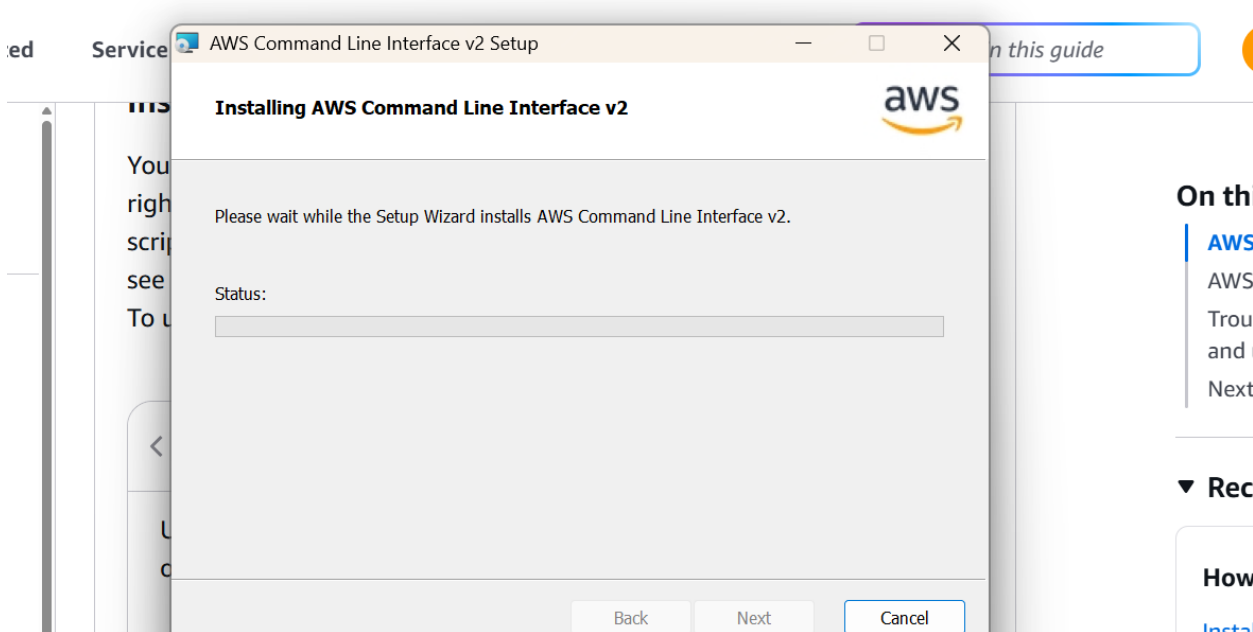
[Cancel](#) [Save](#)

Enable and save it

Launch one EC2 using AWS CLI.



Install command line interface v2



```
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\india>aws --version
aws-cli/2.36.34 Python/3.14.6 Windows/11 exe/AMD64

C:\Users\india>
```


✓ User created successfully



You can view and download the user's password and email instructions for signing in to the AWS Management Console.

[View user](#)

IAM users (1) [Info](#)



[Delete user](#)

[Create user](#)

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

< 1 >

☐	User name	▲	Path	▼	Group: ▼	Last activity	▼	MFA	▼	Password age	▼
☐	ec2-cli-user		/		0	-		-		-	

Set description tag - optional [Info](#)

The description for this access key will be attached to this user as a tag and shown alongside the access key.

Description tag value

Describe the purpose of this access key and where it will be used. A good description will help you rotate this access key confidently later.

Maximum 256 characters. Allowed characters are letters, numbers, spaces representable in UTF-8, and: _ . : / = + - @

[Cancel](#)

[Previous](#)

[Create access key](#)

Retrieve access keys [Info](#)

Access key

If you lose or forget your secret access key, you cannot retrieve it. Instead, create a new access key and make the old key inactive.

Access key

Secret access key

```
Command Prompt
C:\Users\india>aws ec2 run-instances --image-id ami-0c3a3c65a049b6922 --instance-type t3.micro --key-name Amazon-Linux_k
ey --security-group-ids sg-040ealdee8336bc78 --count 1
{
  "ReservationId": "r-0475e2562b3f97f8d",
  "OwnerId": "469132170253",
  "Groups": [],
  "Instances": [
    {
      "Architecture": "x86_64",
      "BlockDeviceMappings": [],
      "ClientToken": "ec8b9684-448e-4281-83f2-2fd1f4ad01b2",
      "EbsOptimized": false,
      "EnaSupport": true,
      "Hypervisor": "xen",
      "NetworkInterfaces": [
        {
          "Attachment": {
            "AttachTime": "2026-08-31T19:53:18+00:00",
            "AttachmentId": "eni-attach-03797562bf3474dc7",
            "DeleteOnTermination": true,
            "DeviceIndex": 0,
            "Status": "attaching",
            "NetworkCardIndex": 0
          },
          "Description": "",
          "Groups": [
            {
              "GroupId": "sg-040ealdee8336bc78",
              "GroupName": "launch-wizard-6"
            }
          ]
        }
      ]
    }
  ]
}
```